

Use Case 1: View Leaderboard

Primary Actor: Player

Secondary Actor: Leaderboard System Actor (the game server)

Purpose of this Use Case: To display the top N scores for a selected game.

Preconditions:

The user (player) has chosen a valid gameId,
and leaderboard data exists (it can be empty).

Main Success Scenario:

1. The Player requests to view the leaderboard for a game.
2. The system sends a request to the game server for the top 'N' scores for the specific gameId.
3. The game server retrieves the top 'N' score entries and sorts them from highest to lowest (not sure if we're going to display all or a top 10 yet).
4. The system displays the leaderboard results back to the player to view.

Extensions of this Use Case:

- Filter Scores is chosen
 - The player can select different variations of sorting the leaderboard: all time, this week, sort order [highest to lowest, lowest to highest], etc.
- No Scores Exist
 - The game server returns an empty list, and the system displays "No Scores Available."

Postconditions:

The leaderboard is displayed,
and no data has been modified

Use Case 2: View My Score

Primary Actor: Player

Secondary Actor: Leaderboard System Actor (the game server)

Purpose of this Use Case: To display the user/player's personal best or all-time scores for a given gameId.

Preconditions:

The player is authenticated (Authenticate User use case <<include>>),
and the player has a valid userID.

Main Success Scenario:

1. The user/player requests to view their score for a specific gameId
2. The system performs Authenticate User (<<include>>)
3. The system sends a request to the game server for the user's top score(s) for a given userID and gameId
4. The game server retrieves the data and displays the result to the player

Extensions of this Use Case:

- Authentication Error/Failure:
 - The system denies access and prompts the player/user to log in again
- The player has no preexisting scores for the game:
 - The game server will return no score
 - The game will display a message "No scores inputted/submitted yet!"

Postconditions:

The player's score information is displayed,
and none of the data has changed

Use Case 3: Submit Score

Primary Actor: Player

Secondary Actor: Leaderboard System Actor (the game server)

Purpose of this Use Case: To submit a new score for the player and update the leaderboard data as well.

Preconditions:

The player has been authenticated,
and the player has a valid userID and chosen gameId,
and there is a valid score value, and it is available to submit

Main Success Scenario:

1. The player submits a new score for a specific gameId
2. The system will perform Authenticate User
3. The system will perform Validate Score Submission
4. The system will send the score submission request to the game server along with a userID, gameId, and score
5. The game server will store the new entry
6. The system will confirm the success of the submission to the user

Extensions of this Use Case:

- Authentication fails
 - The system fails, and the user will be prompted to reauthenticate
- Score validation fails
 - The system will reject the submission and display an error message to the user
- Server / Database error:
 - The system will display a failure message (either prompt retry or automatically attempt again) (not the users fault error)
- Update MMR:
 - After storing the score, the matchmaking system will be updated based on a user's performance (score)

Postconditions:

On a successful attempt, the score is stored and the leaderboard data is updated,
and on a failed attempt, no score will be stored

Authenticate User - Use Case

Goal: Verify the player's identity before moving into the program.

Success: The player is authenticated and can proceed with the program.

Failure: The player's access is denied, and they will be re-prompted to log in.

Validate Score Submission - Use Case

Goal: This ensures a score submission is valid (valid gameId, userID, etc.).

Success: The score is approved and can move into the database and leaderboard.

Failure: The score is denied and prompted with an error message